1

a) false

b) true

c) true

d) true

e) true

2

a) $p \wedge q$

b) $p \wedge \neg q$

c) $\neg p \wedge \neg q$

d) $p \vee q$

e) $p \rightarrow q$

f) $(p \vee q) \wedge p \rightarrow \neg q$

g) $p \leftrightarrow q$

3

a) $2^1 = 2$

b) $2^4 = 16$

c) $2^6 = 64$

d) $2^4 = 16$

4

a)

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

b)

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

c)

p	q	$\neg q$	$p \vee \neg q$	$p \vee \neg q \rightarrow q$
T	T	F	T	T
T	F	T	T	F
F	T	F	F	T
F	F	T	T	F

d)

p	q	$p \vee q$	$p \wedge q$	$(p \vee q) \rightarrow (p \wedge q)$
T	T	T	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T

e)

p	q	$\neg p$	$\neg q$	$p \to q$	$\neg q \to \neg p$	$(p \to q) \leftrightarrow (\neg q \to \neg p)$
T	T	F	F	T	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

f)

p	q	$p \to q$	$q \to p$	$(p \to q) \to (q \to p)$
T	T	T	T	T
T	F	F	T	T
F	T	T	F	F
F	F	T	T	T

(5) "Does everyone want coffee?"

professor #1	professor #2	professor #3
T V F	T V F	F
(yes) (no)	(yes) (no)	(no)

conclusion, first two professors want coffee.

(6) a) Jan is not rich, or Jan is not happy

b) Carlos won't bicycle, and Carlos won't run tomorrow.

c) Mei doesn't walk, and Mei doesn't take the bus to class.

d) Ibrahim is not smart, or Ibrahim is not hard working.

(7) a & b)

p	q	$p \wedge q$	$(p \wedge q) \to p$	$p \to (p \vee q)$	$p \vee q$
T	T	T	T	T	T
T	F	F	T	T	T
F	T	F	T	T	T
F	F	F	T	T	F

c & d)

p	q	$\neg p$	$p \to q$	$\neg p \to (p \to q)$	$p \land q$	$(p \land q) \to (p \to q)$
T	T	F	T	T	T	T
T	F	F	F	T	F	T
F	T	T	T	T	F	T
F	F	T	T	T	F	T

e & f)

p	q	$\neg(p \to q)$	$\neg(p \to q) \to p$	$\neg q$	$\neg(p \to q) \to \neg q$
T	T	F	T	F	T
T	F	T	T	T	T
F	T	F	T	F	T
F	F	F	T	T	T

8

a) $(p \land q) \to p \equiv \neg(p \land q) \lor p$
$\equiv (\neg p \lor \neg q) \lor p$
$\equiv (\neg p \lor p) \lor \neg q$
$\equiv T \lor \neg q$
$\equiv T$

b) $p \to (p \lor q) \equiv \neg p \lor (p \lor q)$
$\equiv (\neg p \lor p) \lor q$
$\equiv T \lor q$
$\equiv T$

c) $\neg p \to (p \to q) \equiv \neg(\neg p) \lor (\neg p \lor q)$
$\equiv (p \lor \neg p) \lor q$
$\equiv T \lor q$
$\equiv T$

d) $(p \land q) \to (p \to q) \equiv$
$\neg(p \land q) \lor (\neg p \lor q) \equiv$
$\neg p \lor \neg q \lor \neg p \lor q \equiv$
$(\neg p \lor \neg p) \lor (\neg q \lor q) \equiv$
$F \lor T \equiv$
T

e) $\neg(p \rightarrow q) \rightarrow p \equiv \neg(\neg(p \rightarrow q)) \vee p$
$\equiv \neg(\neg(\neg p \vee q)) \vee p$
$\equiv \neg(p \wedge \neg q) \vee p$
$\equiv \neg p \vee q \vee p$
$\equiv \neg p \vee p \vee q$
$\equiv T \vee q$
$\equiv T$

f) $\neg(p \rightarrow q) \rightarrow \neg q \equiv$
$\neg(\neg(p \rightarrow q)) \vee \neg q \equiv$
$\neg(\neg(\neg p \vee q)) \vee \neg q \equiv$
$\neg(p \wedge \neg q) \vee \neg q \equiv$
$\neg p \vee (q \vee \neg q) \equiv$
$\neg p \vee T \equiv$
T

9. $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (\neg p \vee r) \wedge (\neg q \vee r)$
$\equiv (r \vee \neg p) \wedge (r \vee \neg q)$
$\equiv r \vee (\neg p \wedge \neg q)$
$\equiv r \vee \neg(p \vee q)$
$\equiv \neg(p \vee q) \vee r \equiv (p \vee q) \rightarrow r$

10.
a) $\exists x\,(P(x) \wedge Q(x))$
b) $\exists x\,(P(x) \wedge \neg Q(x))$
c) $\forall x\,(P(x) \vee Q(x))$
d) $\forall x\,((\neg P(x)) \wedge (\neg Q(x)))$

11.
a) $\forall n\,(n^2 \geq 0)$ TRUE
b) $\exists n\,(n^2 = 2)$ FALSE
c) $\forall n\,(n^2 \geq n)$ TRUE
d) $\exists n\,(n^2 < 0)$ FALSE

(12)

a) $P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4)$

b) $P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4)$

c) $\neg P(0) \vee \neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$

d) $\neg P(0) \wedge \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4)$

e) $\neg P(0) \wedge \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4)$

f) $\neg P(0) \vee \neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$

(13)

a) $\exists_x H(x)$, $\exists_x (C(x) \wedge H(x))$

b) $\forall_x f(x)$, $\forall_x (C(x) \to f(x))$

c) $\exists_x \neg B(x)$, $\exists_x (C(x) \wedge \neg B(x))$

d) $\exists_x M(x)$, $\exists_x (C(x) \wedge M(x))$

e) $\forall_x \neg P(x)$, $\forall_x (C(x) \to \neg P(x))$

(14)

a) $\neg M(\text{Chace}, \text{Koko})$

b) $\neg M(\text{Arlene}, \text{Sarah}) \wedge \neg T(\text{Arlene}, \text{Sarah})$

c) $\neg M(\text{Deborah}, \text{Jose})$

d) $\forall_x M(x, \text{Ken})$

e) $\neg \exists_x T(x, \text{Nina})$

f) $\forall_x (T(x, \text{Avi}) \vee M(x, \text{Avi}))$

g) $\exists_x \forall_y \, (y \neq x \to M(x,y))$

h) $\exists_x \forall_y \, (y \neq x \to (M(x,y) \vee T(x,y)))$

i) $\exists_x \exists_y (x \neq y \wedge M(x,y) \wedge M(y,x))$

j) $\exists_x M(x,x)$

k) $\exists_x \forall_y (x \neq y \to (\neg M(y,x) \wedge \neg T(y,x)))$

l) $\forall_x \exists_y \, (x \neq y \wedge (M(y,x) \vee T(y,x)))$

m) $\exists_x \exists_y \, (x \neq y \wedge M(x,y) \wedge T(y,x))$

n) $\exists_x \exists_y \, (x \neq y \wedge \forall_z ((z \neq x \wedge z \neq y) \to (M(x,z) \vee M(y,z) \vee T(x,z) \vee T(y,z))))$